

Ethanol Blending in Gasoline – Czechia

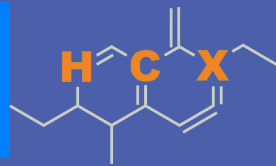
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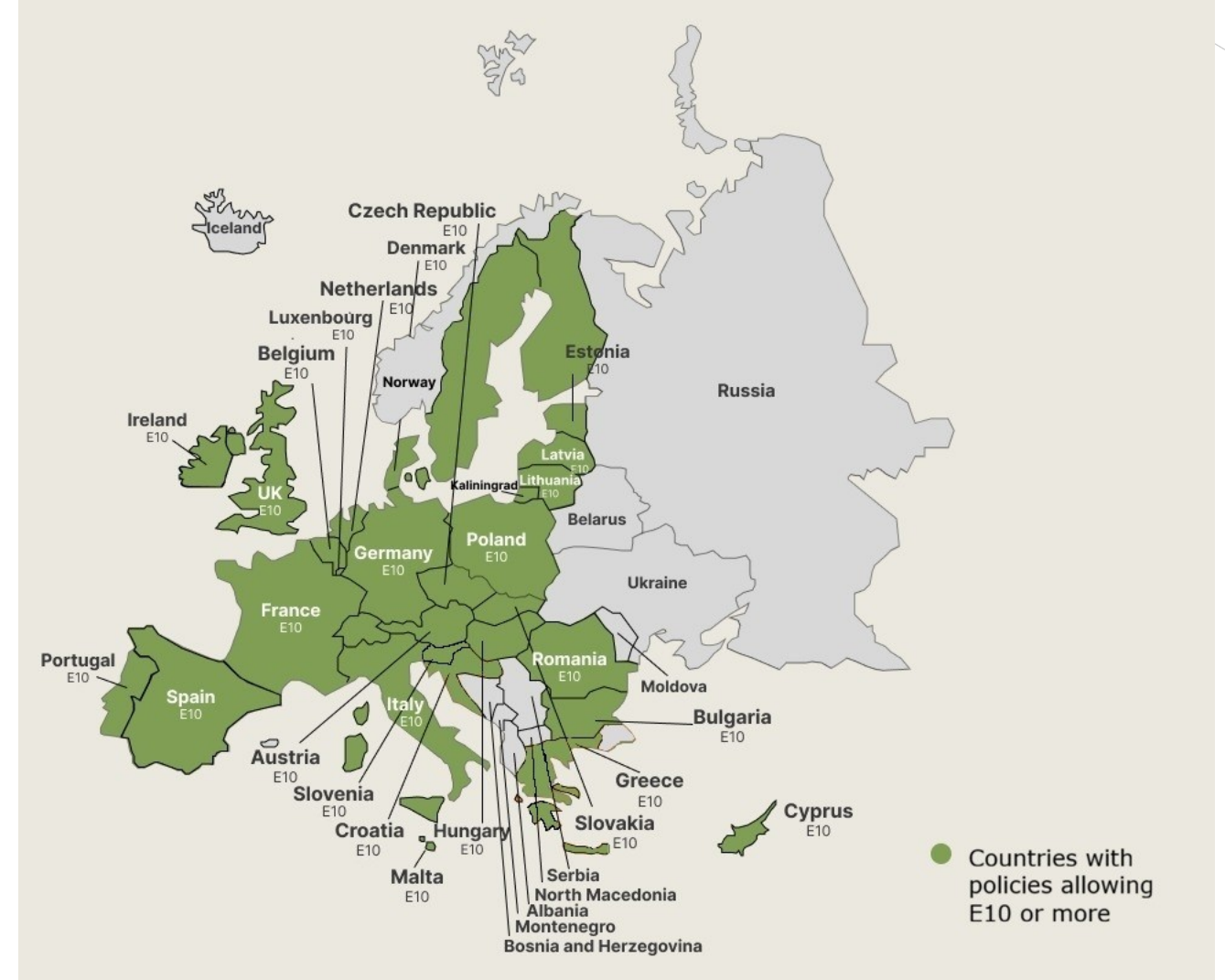


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Ethanol Blending in Europe

There are important fuel quality and environmental impact of vehicle emission challenges in the Region.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Europe to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.



Twenty eight countries with potential and additional use of ethanol were studied: 1) Gasoline market profiles; 2) Optimization of gasoline blends with ethanol and, 3) Environmental impact of gasolines blended with ethanol.

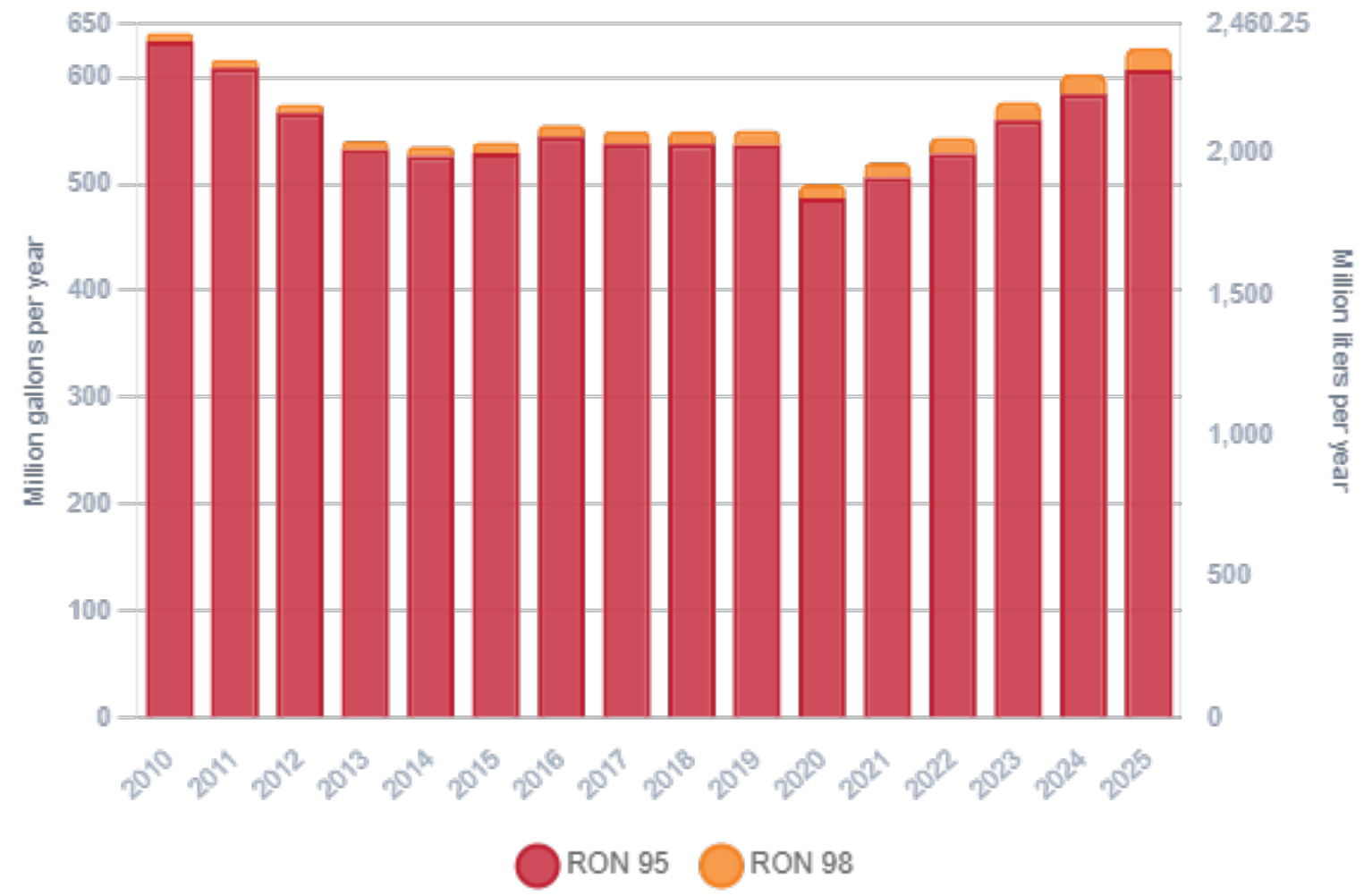
Ethanol Gasoline Blending in Czechia



In 2025, gasoline consumption in Czechia was 2,445 million liters (646 million gallons) and growth rate was -0.19%. Gasoline production and net imports were 2,033 and 214 million liters (537 and 56 million gallons) correspondingly. Czechia complies with the European gasoline standards and offers two main grades: RON 95 and RON 98 with an average octane of 95.1 RON.

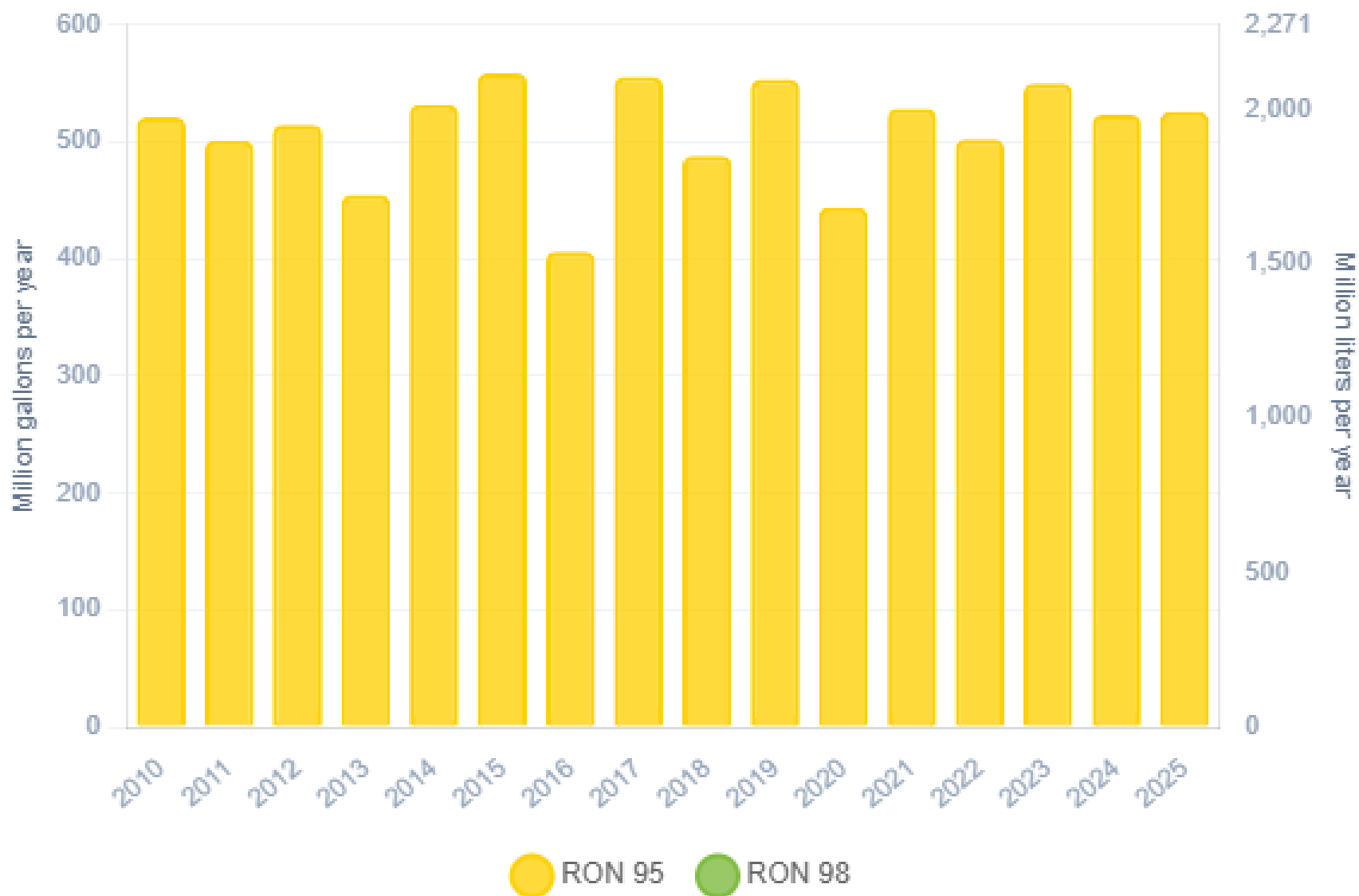
In 2025, ethanol consumption in Czechia was 227 million liters (60 million gallons) representing 10% of gasoline demand. Ethanol production and Net Imports were 47 and 123 million liters (12 and 32 million gallons) correspondingly. Ethanol sources are Corn 50% and Sugar Beet 50%.

Gasoline Demand in Czechia



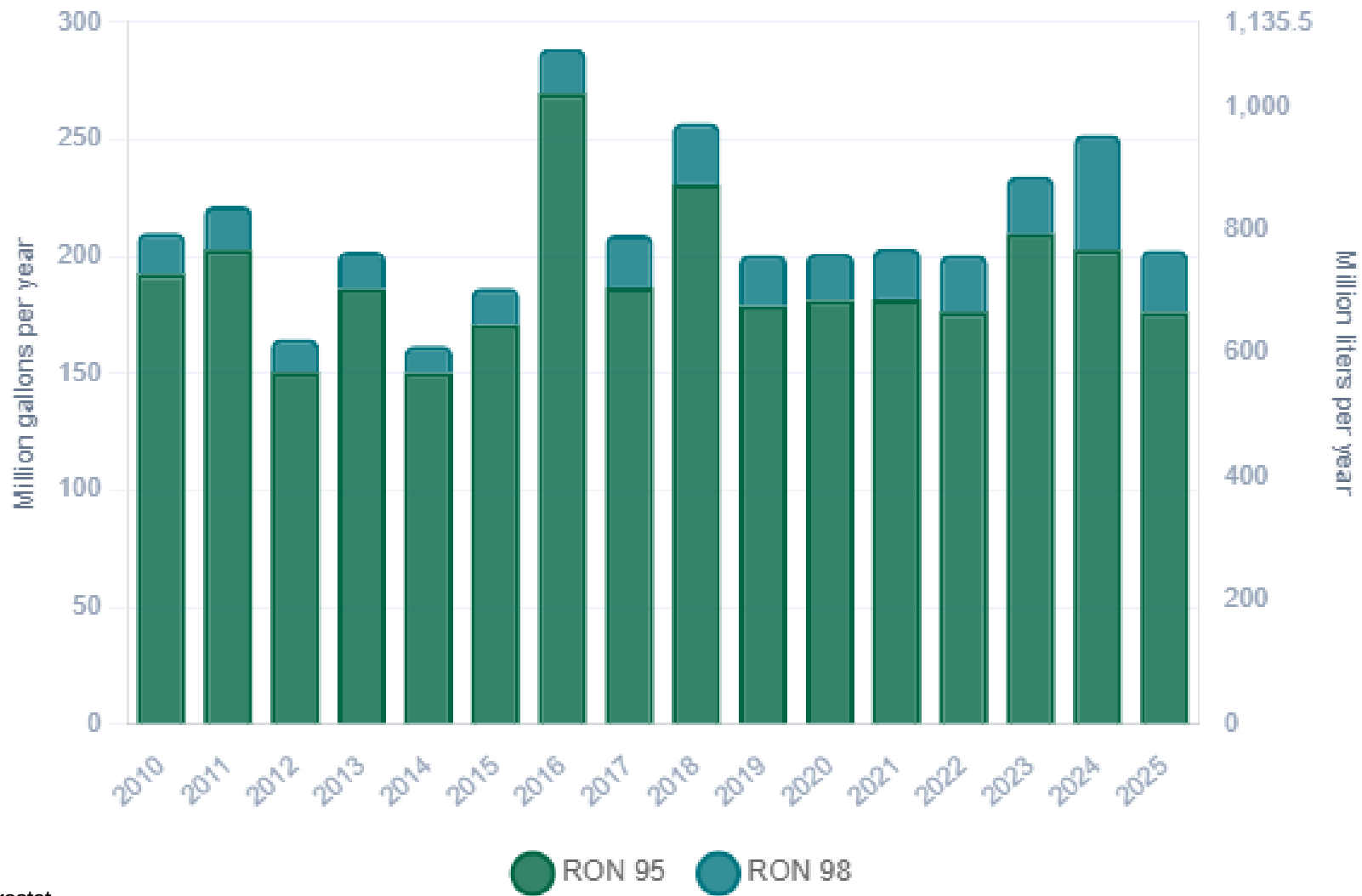
Source: Eurostat

Gasoline Production in Czechia



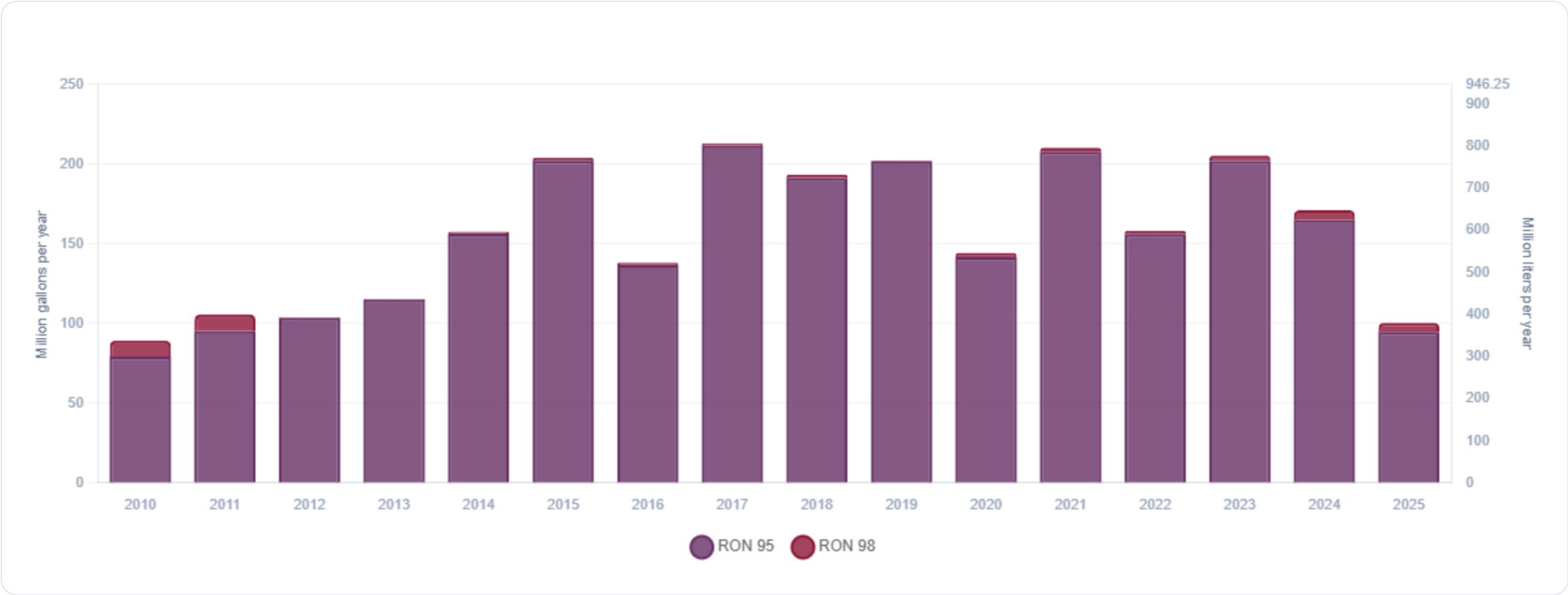
Source: Eurostat

Gasoline Imports in Czechia



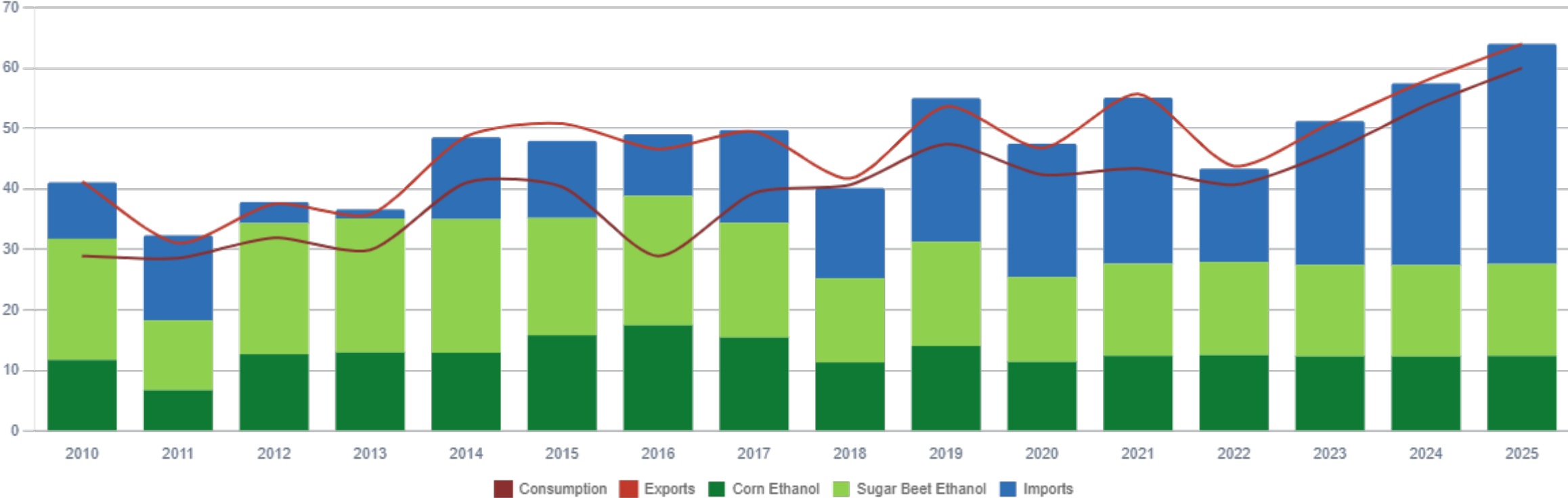
Source: Eurostat

Gasoline Exports in Czechia



Source: Eurostat

Ethanol Balance in Czechia



Source: Eurostat

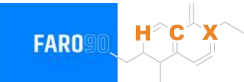
Gasoline Quality in Czechia

Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Countries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	-							
Advanced Biofuels (%cal)	1.1							
Bioethanol in Gasoline Sales (%cal)	-							
GHG Emission Reduction (%)	6.0							

* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

Source: European Commission

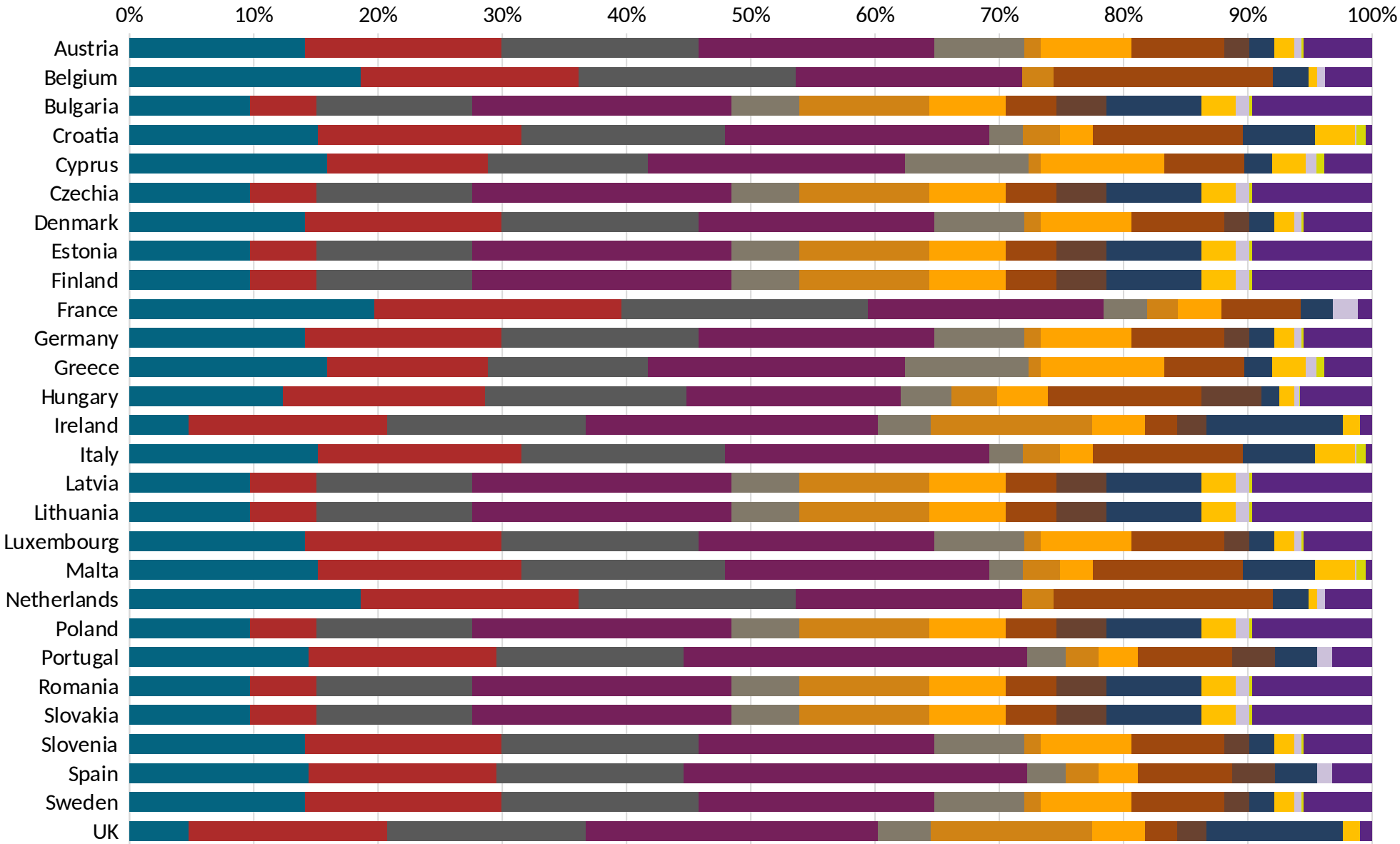
Gasoline Component Blending in Europe



Gasoline is a blend of a base gasoline and other components. Each component provides different properties to the final blend, for example, isomerates, alkylates and butanes increase the octane. The components commonly used in Europe are:

- Catalytic Gasoline
- Light Primary Naphta
- Heavy Primary Naphta
- Reformate
- Isopentane
- Alkylate
- Isomerate
- Hydrocracked Gasoline
- Coker Naphta
- n-Butane
- MTBE
- ETBE
- TAME
- Aromatics

Source:
Faro90



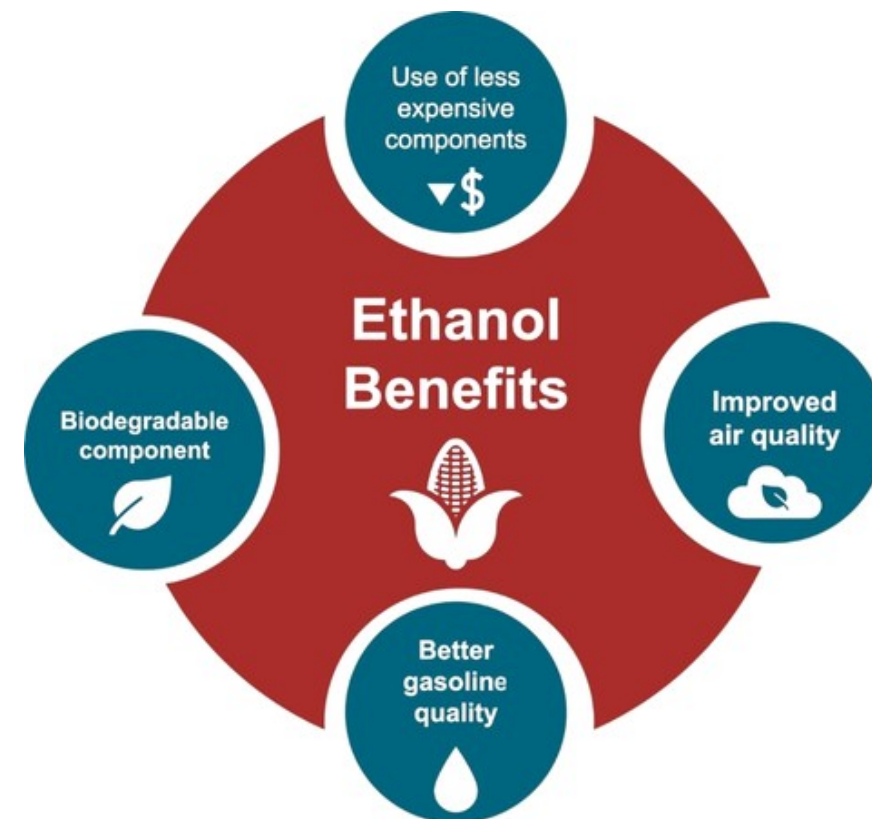
Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component. The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives. To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

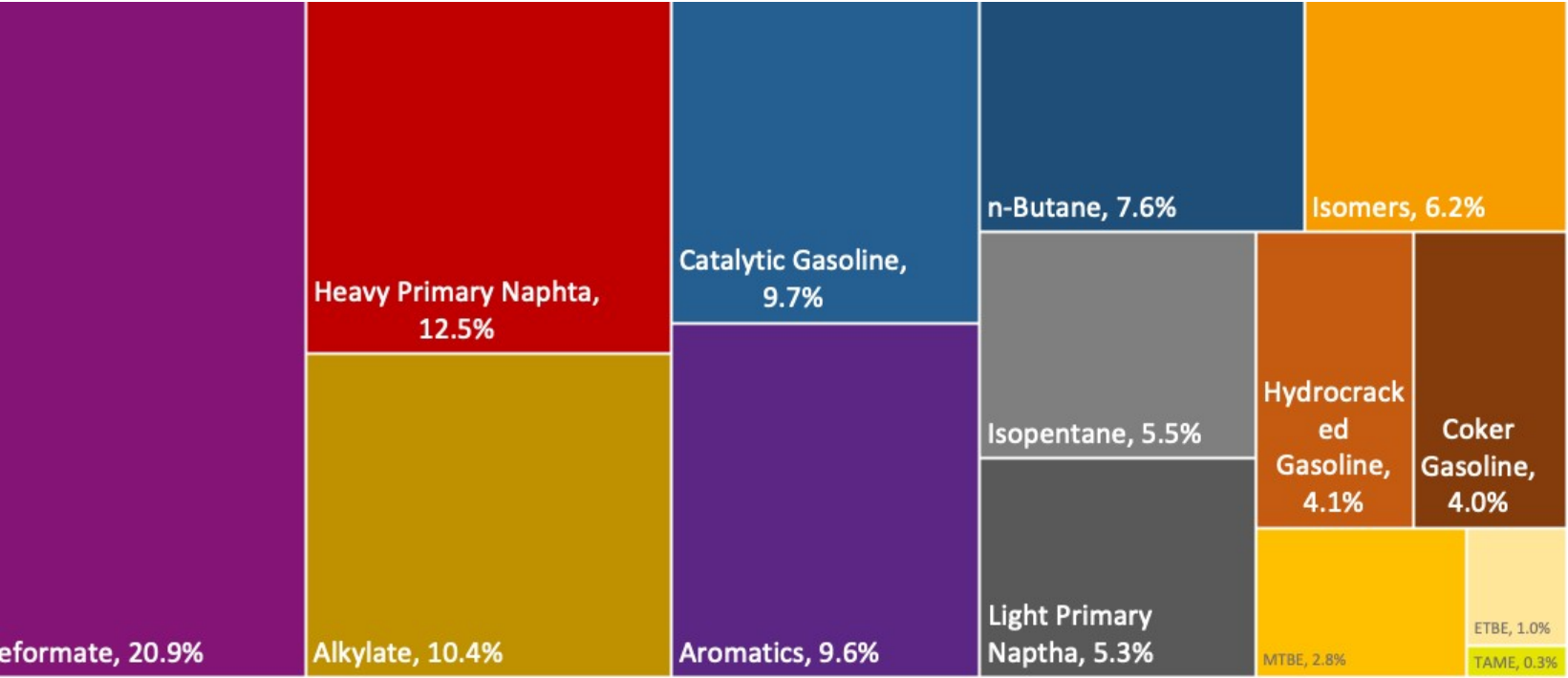
- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

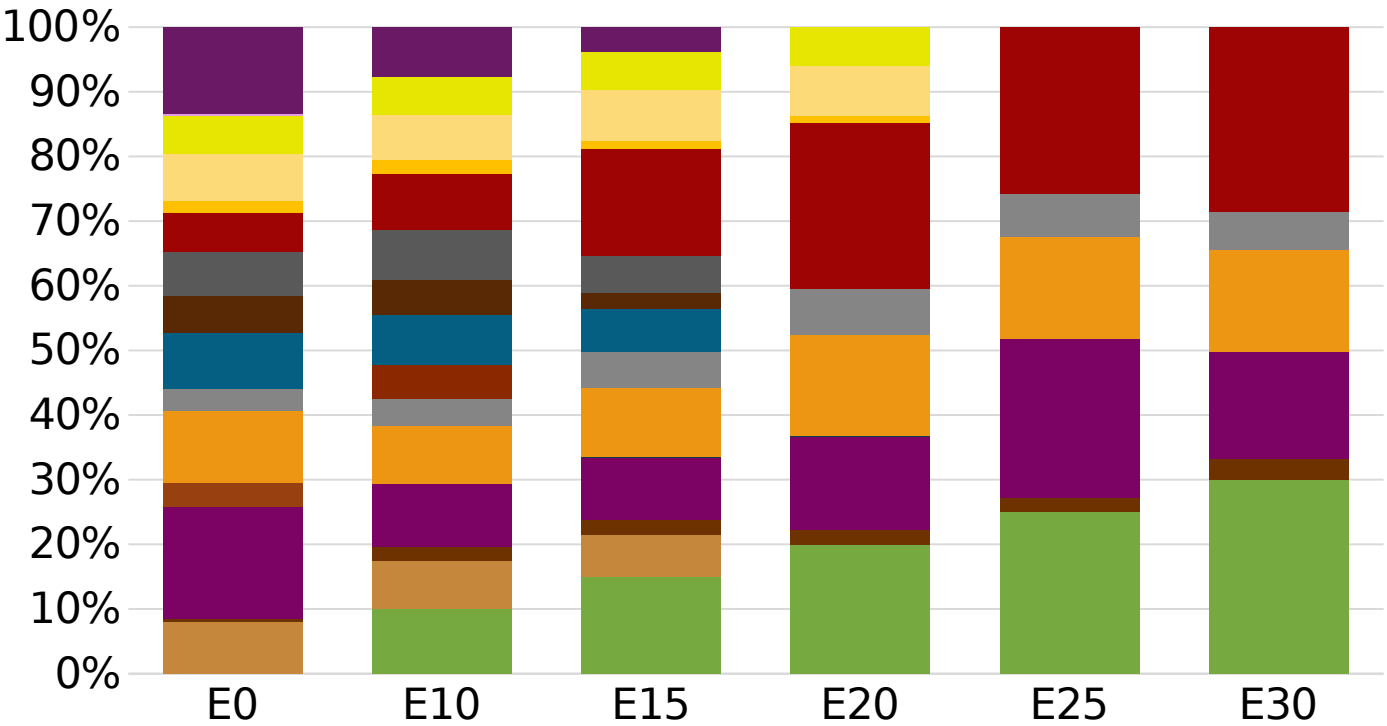
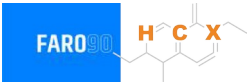
The blending model uses gasoline component spot average prices 2025 and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.



Czechia Available Blending Components



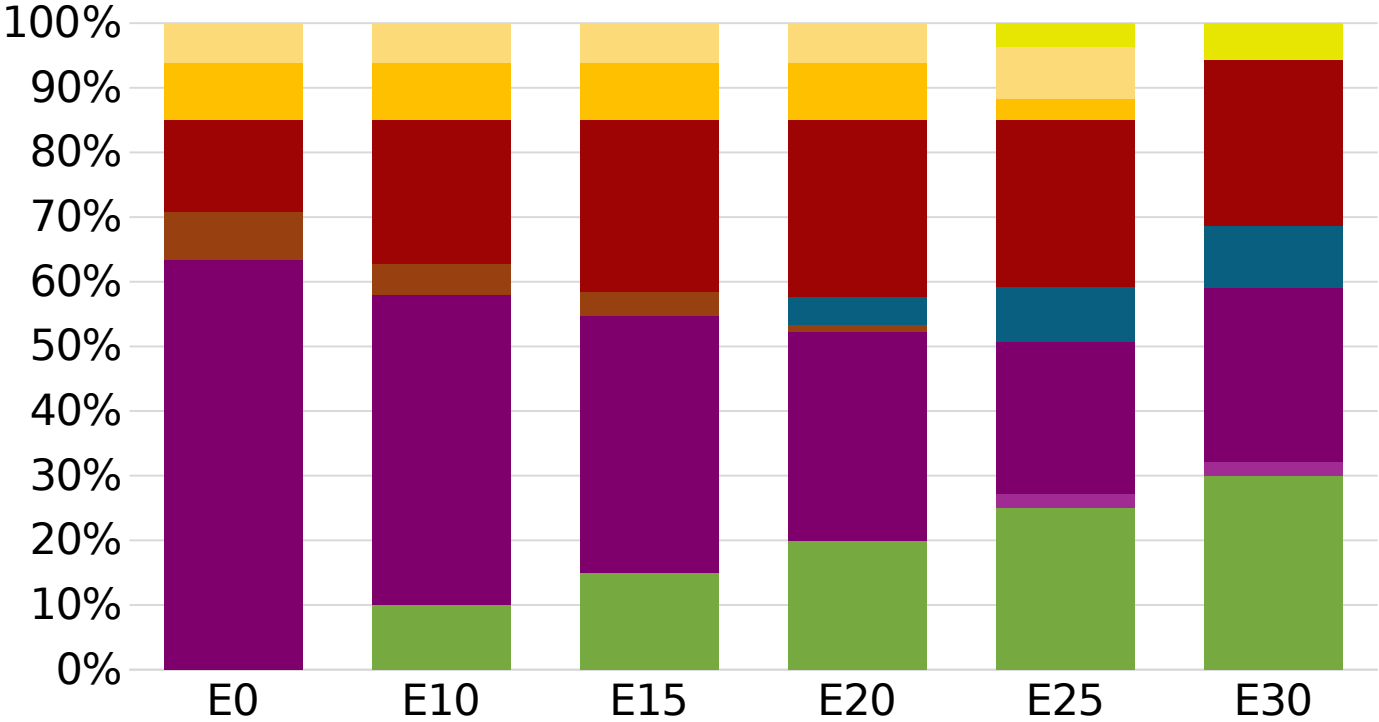
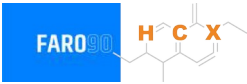
Czechia – Gasoline 95 – Constant Octane



Average CIF 2025
Prices

Octane (RON)	95.0	95.0	95.0	95.0	95.0	95.8
Price (US\$/gal)	\$ 2.111	\$ 1.973	\$ 1.873	\$ 1.758	\$ 1.712	\$ 1.659

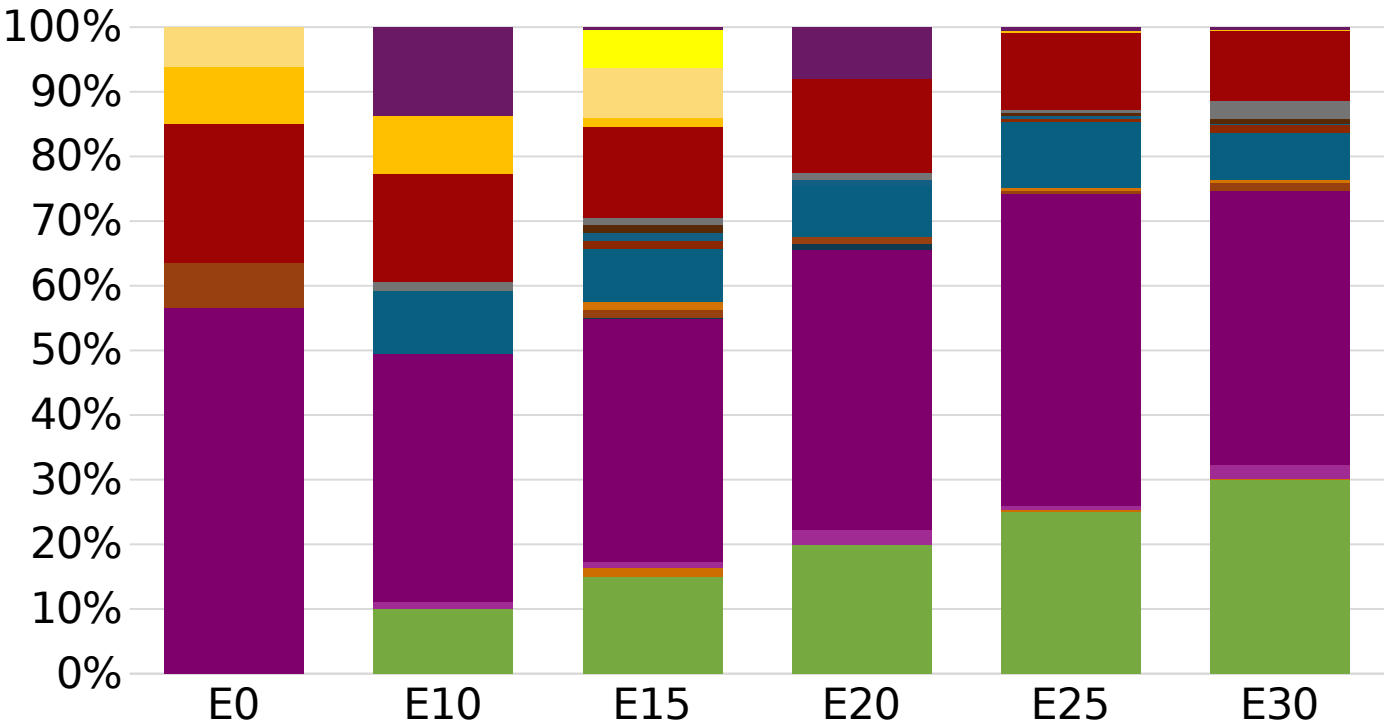
Czechia - Gasoline 98 - Constant Octane



Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US\$/gal)	\$ 2.131	\$ 2.014	\$ 1.945	\$ 1.873	\$ 1.808	\$ 1.758

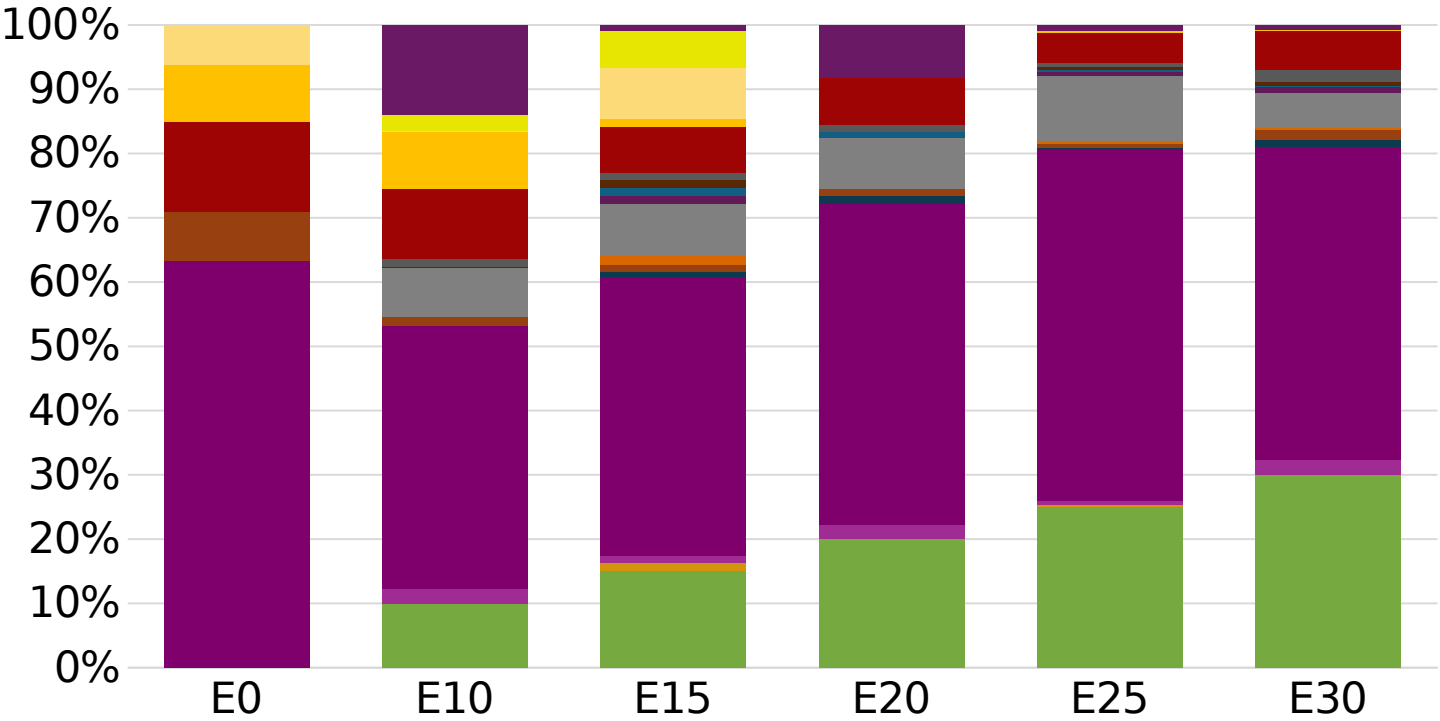
Average CIF 2025
Prices

Czechia – Gasoline 95 – Increased Octane



Average CIF 2025
Prices

Czechia – Gasoline 98 – Increased Octane



Average CIF 2025
Prices

Octane (RON)	98.0	99.0	100.9	101.2	102.8	104.1
Price (US\$/gal)	\$ 2.131	\$ 2.131	\$ 2.131	\$ 2.131	\$ 2.131	\$ 2.131

Vehicle Emission Impact for Ethanol Gasoline Blending

The model used in this analysis takes as a reference the [International Vehicle Emissions Model \(IVE\)](#).

The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

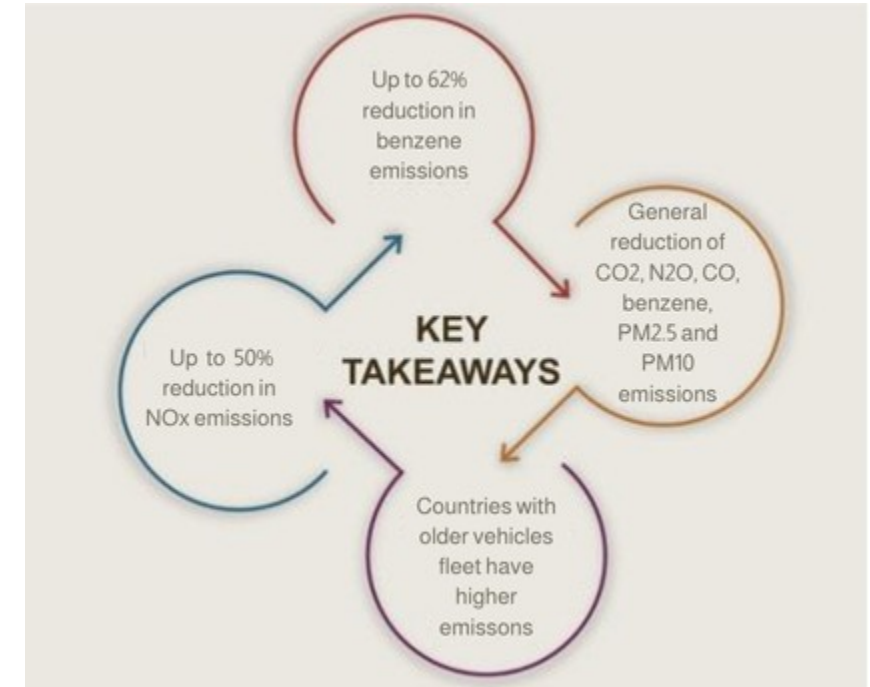
- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

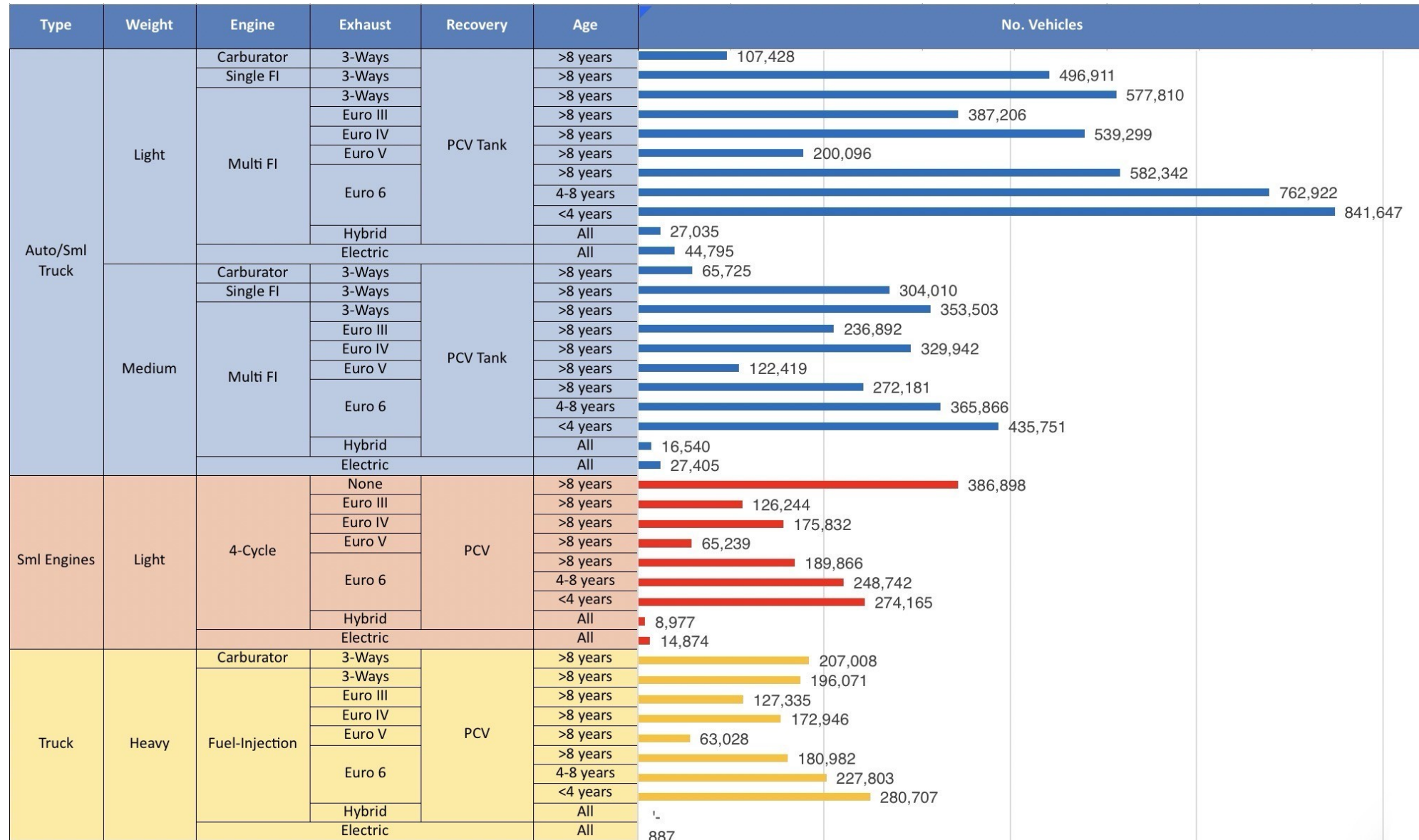
Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

**Source: Bureau of transportation statistics.*

Main Results



Gasoline Vehicle Fleet - Czechia

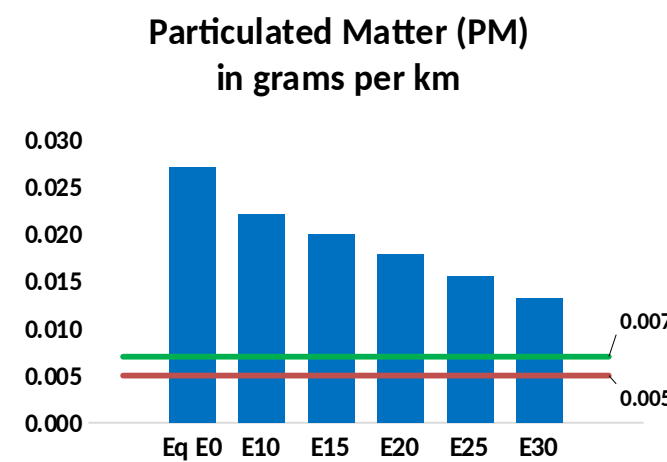
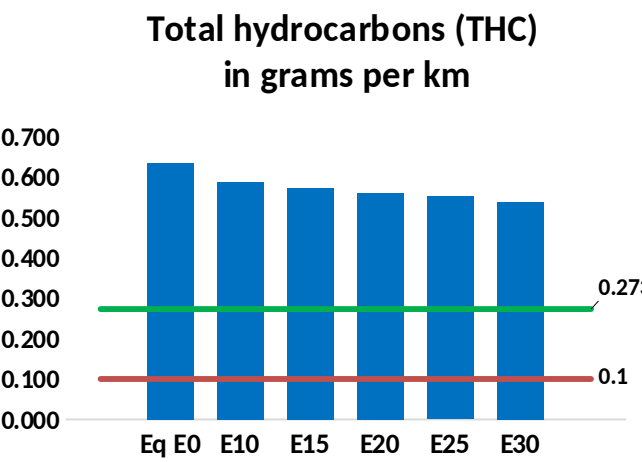
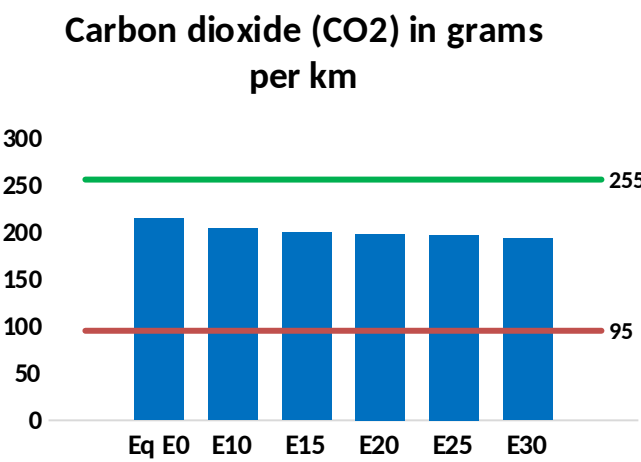
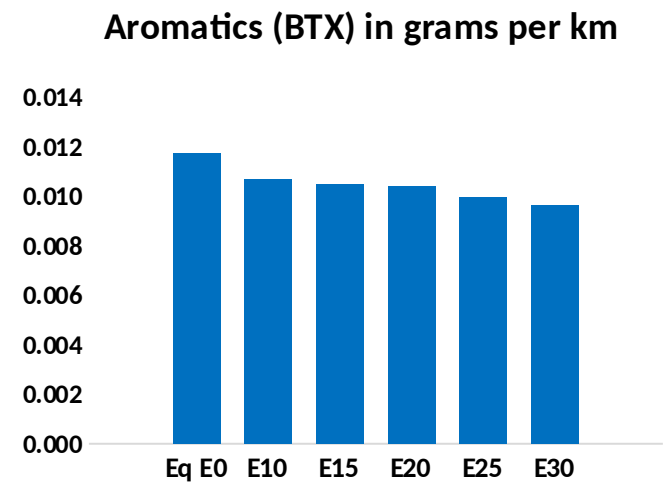
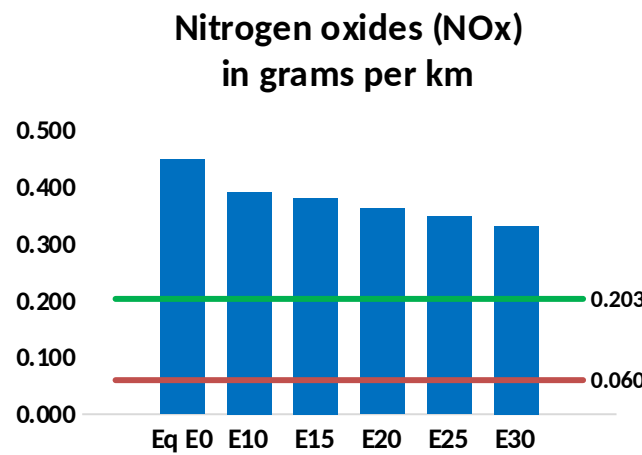
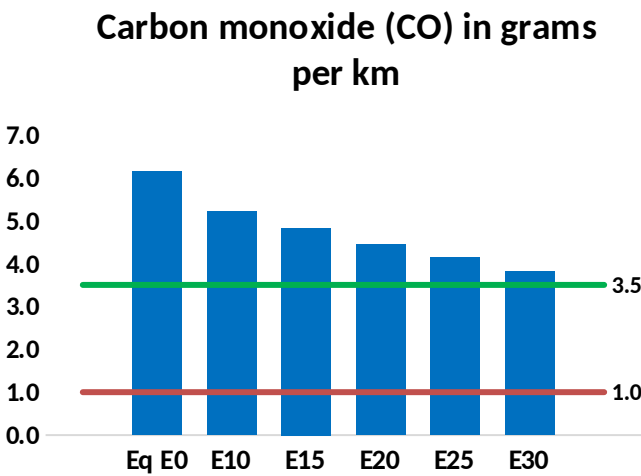
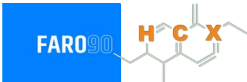


Vehicle Fleet: 10,045,330
Gasoline Vehicle Fleet: 3,692,443

Source: Eurostat, ACEA

Czechia – Vehicular Emissions

TIER III BIN 125 United States
Euro 6 Standard



Czechia – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	521,318	481,576	441,834	409,332	376,655	351,537	322,496	-15%	-28%
CO2	18,149,222	17,682,528	17,241,761	16,895,111	16,723,575	16,642,885	16,336,109	-5%	-8%
VOC	53,728	51,740	49,753	48,543	47,517	46,786	45,607	-7%	-12%
NOx	38,011	35,097	33,189	32,265	30,723	29,538	28,129	-13%	-19%
SOx	208	193	177	164	151	137	122	-15%	-28%
NH3	6,238	6,176	6,115	6,144	6,213	6,261	6,302	-2%	0%
N2O	317	315	314	316	322	326	329	-1%	2%
CH4	11,854	11,854	11,854	12,032	12,292	12,518	12,731	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	362	344	326	313	301	292	279	-10%	-17%
Acetaldehyde	765	878	1,288	1,961	2,672	3,053	3,608	68%	249%
Formaldehyde	2,499	2,429	2,832	3,325	3,484	3,854	4,195	13%	39%
Benzene	994	955	905	891	884	845	818	-9%	-11%
PM2.5	1,040	945	851	768	686	599	508	-18%	-34%
PM10	1,254	1,140	1,026	926	828	722	612	-18%	-34%

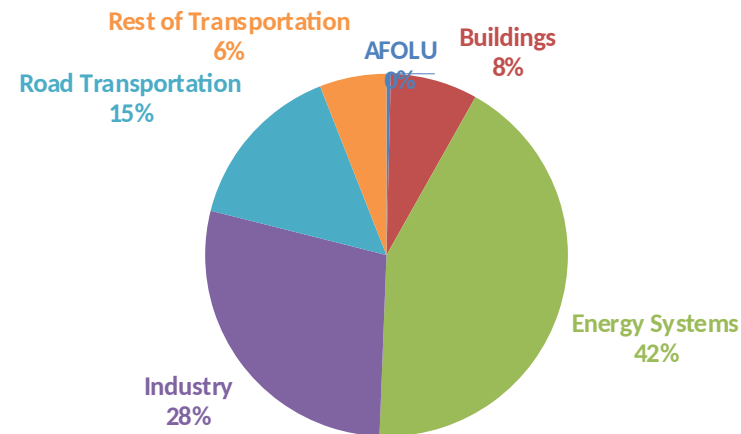
Life Cycle GHG Emissions

The Global Agenda for Climate Action calls for countries, cities and regions, businesses, and members of civil society around the world to achieve a low-carbon and climate-resilient economies in support of the Paris Agreement. Road Transportation accounts for 15% of total GHG emissions (IPCC,2023).

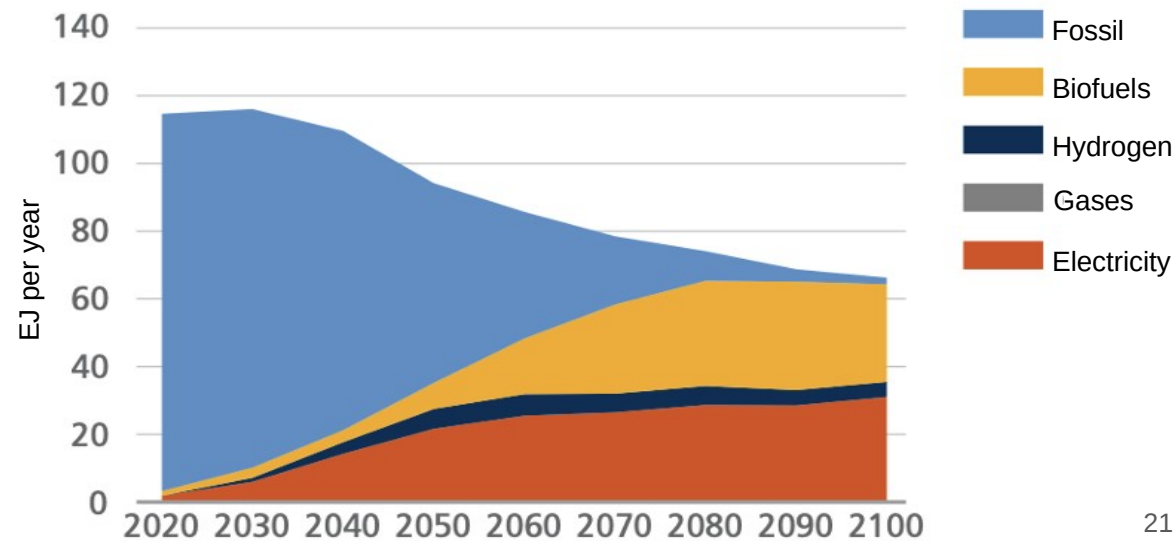
Ethanol gasoline blends are an internationally adopted practice that offer significant short-term benefits. Biofuels are an important solution to achieve reduction targets in the transportation sector.

Country GHG emission reductions are calculated using **RED II/III** and **GREET** life cycle emission methodologies and the **International Vehicular Emissions Model (IVE)**. Results are compared to the current country targets for 2035. Current 2025 country emission are those published by **IPCC**.

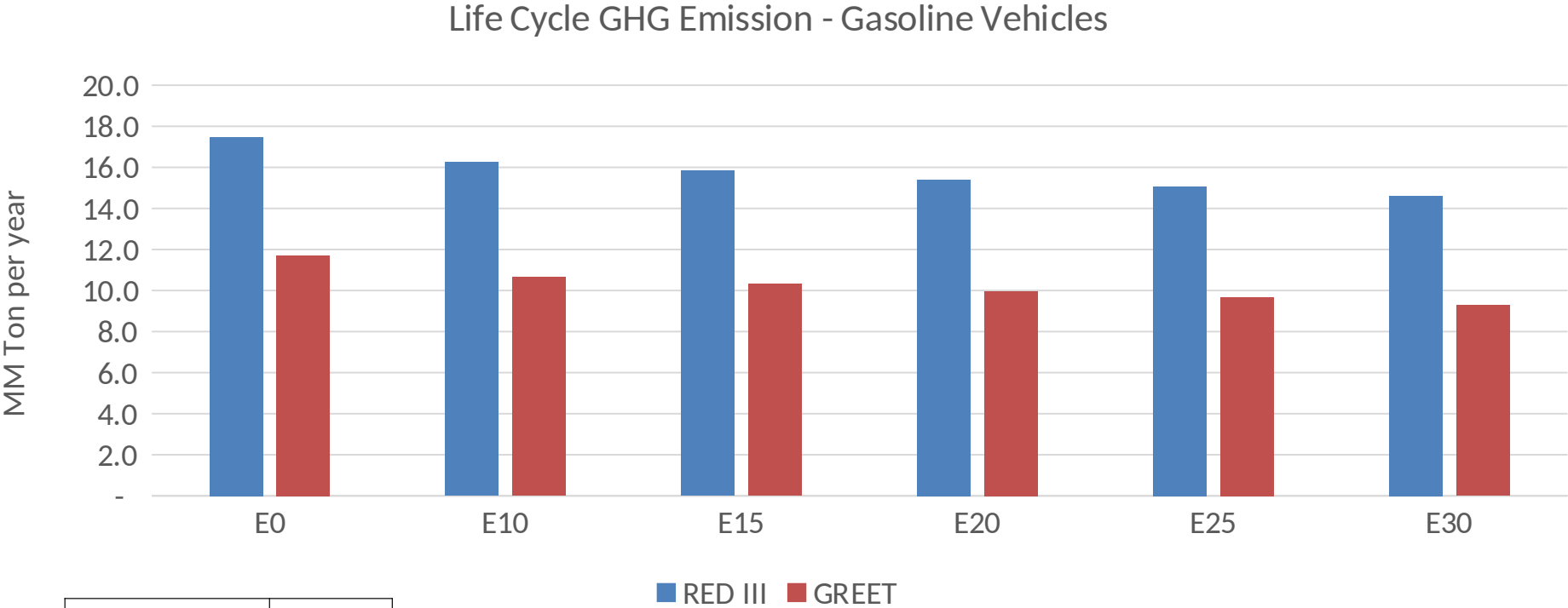
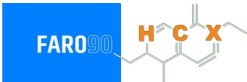
Current GHG Global Emissions Distribution



IPCC Transport Sector Sustainable Development Goals and climate policies Scenario, 2023



Czechia – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2025 (MMT/año)	108.0
Target @ 2035 (MMT/year)	60.8
Delta	47.2